

Evolutionary Computation and its Applications

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Today's goals



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Module 1

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Fundamental Concepts in Optimization

Welcome

- Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min
- Specialization and Course Structure

Reading • 10 min
- Specialization Sample Certificate

Reading • 10 min
- ✓

Stay connected with UofG Online

Reading • 5 min
- ✓

Why Optimization Matters in Engineering and Science

Reading • 5 min

Module 4 Introduction

- ✓

Introduction to Module 4's Study

Video • 1 min
- ✓

Optimization and Evolutionary Computation in AI

Reading • 5 min

Module 4-1 Introduction to Optimization

- Introduction to Optimization: Concepts and Applications

Video • 10 min
- Introduction to Optimization: Further Applications

Video • 6 min
- Materials about Optimization

Reading • 10 min

Module 4-2 Optimization Objectives and Constraints

- Optimization Objectives and Constraints: Objectives and Constraints

Video • 6 min
- Optimization Objectives and Constraints: Multiple Objectives

Video • 9 min
- Materials about Optimization Objectives and Constraints

Reading • 10 min

Module 4-3 Global and Local Optimization

- Global and Local Optimization: Concepts

Video • 9 min
- Materials about Global and Local Optimization

Reading • 10 min

Module 4-4 Formulate Engineering Problems to Optimization Problems

- Formulate Engineering Problems to Optimization Problems: Procedures and A Case Study

Video • 11 min
- Materials about Optimization Problem Formulation

Reading • 10 min

Module 4-5 Case Studies

- Case Studies (1)

Video • 3 min
- Case Studies (2)

Video • 7 min
- Case Studies (3)

Video • 3 min
- Case Studies (4)

Video • 45 min

Optimization and Evolutionary Computation in AI

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In this section, you reflected on the role of optimization within the broader landscape of AI, particularly beyond the narrow view of AI as only machine learning.

Now, reflect on your own understanding:

- Why is optimization fundamentally important in your field? Where are the real decisions being made, and how often do they reduce to finding the best configuration under constraints rather than simply building predictive models?
- Have you been limiting your view of AI to machine learning? What other branches exist—such as search, planning, knowledge-based systems, and optimization—and how do they differ in their purpose and practical application?
- Where does evolutionary computation fit within this landscape? How is it different from data-driven learning methods, particularly in terms of how solutions are generated, evaluated, and improved?
- What specific evolutionary computation techniques do you know (e.g., genetic algorithms, genetic programming), and what types of problems are they best suited to solving? Are you clear on why they perform well for certain classes of problems, such as complex, non-convex, or poorly understood search spaces?

This reflection encourages you to broaden your perspective of AI beyond machine learning and to better understand the role of optimization-driven approaches in solving real-world engineering and scientific problems.

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✓ Completed

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